House Numbers for Energy Usage

8414	346726	273829
91964	422441	401347
426099	386245	446418
227084	233297	363396
15263	248114	382806
10529	333886	308995
227394	76106	215101
237695	472680	344346
9590	282561	113638
226002	122693	346083
464861	273912	347073
127412	421897	294974
281810	62433	428089
179732	273002	388882
43131	274947	447945
394268	55646	315508
452857	239951	218528
95304	440966	153775
147770	215797	412481
184935	27488	400236
423651	28610	126343
287266	457641	193618
441558	338309	135778
277961	200860	311488
151086	66593	252215
468173	111057	452081
180942	429160	385929
124597	207620	448177
341162	424705	121
162387	17172	67052
396821	64499	317721
183363	266477	77464
69859	156254	74770
350425	385315	224243
379022	472022	69968
320971	301541	66925
456064	114407	438302
223580	296273	180894

316859	24138	110330
394162	353448	187375
292336	111719	408767
120826	213979	380842
186846	442759	5081
410629	362784	312606
75945	352663	15946
46952	297913	224833
130302	176945	152536
205782	164661	319933
440212	351272	213330
382449	121342	421827

County Numbers For Weather

"G4500430"
"G4500830"
"G4500630"
"G4500150"
"G4500870"
"G4500190"
"G4500590"
"G4500530"
"G4500510"
"G4500270"
"G4500790"
"G4500350"
"G4500410"
"G4500070"
"G4500570"
"G4500130"
"G4500030"
"G4500450"
"G4500910"
"G4500710"
"G4500250"
"G4500390"
"G4500550"
"G4500850"
"G4500750"
"G4500230"
"G4500890"
"G4500290"
"G4500210"
"G4500690"
"G4500730"
"G4500670"
"G4500770"
"G4500370"
"G4500470"
"G4500310"

All Data Columns:

“bldg_id"
"time"
"in.county"
"out.electricity.ceiling_fan.energy_consumption"

"out.electricity.clothes_dryer.energy_consumption"
"out.electricity.clothes_washer.energy_consumption"
"out.electricity.cooling_fans_pumps.energy_consumption"
"out.electricity.cooling.energy_consumption"
"out.electricity.dishwasher.energy_consumption"
"out.electricity.freezer.energy_consumption"

"out.electricity.heating_fans_pumps.energy_consumption"
"out.electricity.heating_hp_bkup.energy_consumption"
"out.electricity.heating.energy_consumption"
"out.electricity.hot_tub_heater.energy_consumption"
"out.electricity.hot_tub_pump.energy_consumption"
"out.electricity.hot_water.energy_consumption"
"out.electricity.lighting_exterior.energy_consumption"
"out.electricity.lighting_garage.energy_consumption"

"out.electricity.lighting_interior.energy_consumption"
"out.electricity.mech_vent.energy_consumption"

"out.electricity.plug_loads.energy_consumption"
"out.electricity.pool_heater.energy_consumption"
"out.electricity.pool_pump.energy_consumption"
"out.electricity.pv.energy_consumption"
"out.electricity.range_oven.energy_consumption"
"out.electricity.refrigerator.energy_consumption"
"out.electricity.well_pump.energy_consumption"
"out.fuel_oil.heating_hp_bkup.energy_consumption"
"out.fuel_oil.heating.energy_consumption"
"out.fuel_oil.hot_water.energy_consumption"
"out.natural_gas.clothes_dryer.energy_consumption"
"out.natural_gas.fireplace.energy_consumption"
"out.natural_gas.grill.energy_consumption"
"out.natural_gas.heating_hp_bkup.energy_consumption"
"out.natural_gas.heating.energy_consumption"
"out.natural_gas.hot_tub_heater.energy_consumption"
"out.natural_gas.hot_water.energy_consumption"
"out.natural_gas.lighting.energy_consumption"
"out.natural_gas.pool_heater.energy_consumption"
"out.natural_gas.range_oven.energy_consumption"
"out.propane.clothes_dryer.energy_consumption"
"out.propane.heating_hp_bkup.energy_consumption"
"out.propane.heating.energy_consumption"

"out.propane.hot_water.energy_consumption"
"out.propane.range_oven.energy_consumption"
"in.sqft"
"in.bathroom_spot_vent_hour"
"in.bedrooms"
"in.building_america_climate_zone"
"in.cec_climate_zone"
"in.ceiling_fan"
"in.city"
"in.clothes_dryer"
"in.clothes_washer"
"in.clothes_washer_presence"
"in.cooking_range"
"in.cooling_setpoint"
"in.cooling_setpoint_has_offset"
"in.cooling_setpoint_offset_magnitude"
"in.cooling_setpoint_offset_period"
"in.dehumidifier"
"in.dishwasher"
"in.door_area"
"in.doors"
"in.ducts"
"in.eaves"
"in.electric_vehicle"
"in.emissions_electricity_units"
"in.emissions_fossil_fuel_units"
"in.emissions_fuel_oil_values"
"in.emissions_natural_gas_values"
"in.emissions_propane_values"
"in.emissions_scenario_names"
"in.emissions_types"
"in.emissions_wood_values"
"in.federal_poverty_level"
"in.generation_and_emissions_assessment_region"
"in.geometry_attic_type"
"in.geometry_building_horizontal_location_mf"
"in.geometry_building_horizontal_location_sfa"
"in.geometry_building_level_mf"
"in.geometry_building_number_units_mf"
"in.geometry_building_number_units_sfa"
"in.geometry_building_type_acs"
"in.geometry_building_type_height"
"in.geometry_building_type_recs"
"in.geometry_floor_area"
"in.geometry_floor_area_bin"
"in.geometry_foundation_type"
"in.geometry_garage"
"in.geometry_stories"
"in.geometry_stories_low_rise"
"in.geometry_story_bin"
"in.geometry_wall_exterior_finish"
"in.geometry_wall_type"
"in.has_pv"
"in.heating_fuel"
"in.heating_setpoint"
"in.heating_setpoint_has_offset"
"in.heating_setpoint_offset_magnitude"
"in.heating_setpoint_offset_period"
"in.holiday_lighting"
"in.hot_water_distribution"
"in.hot_water_fixtures"
"in.hvac_cooling_efficiency"
"in.hvac_cooling_partial_space_conditioning"
"in.hvac_cooling_type"
"in.hvac_has_ducts"
"in.hvac_has_shared_system"
"in.hvac_has_zonal_electric_heating"
"in.hvac_heating_efficiency"
"in.hvac_heating_type"
"in.hvac_heating_type_and_fuel"
"in.hvac_secondary_heating_efficiency"
"in.hvac_secondary_heating_type_and_fuel"
"in.hvac_shared_efficiencies"

"in.hvac_system_is_faulted"
"in.hvac_system_single_speed_ac_airflow"
"in.hvac_system_single_speed_ac_charge"
"in.hvac_system_single_speed_ashp_airflow"
"in.hvac_system_single_speed_ashp_charge"
"in.income"
"in.income_recs_2015"
"in.income_recs_2020"
"in.infiltration"
"in.insulation_ceiling"
"in.insulation_floor"
"in.insulation_foundation_wall"
"in.insulation_rim_joist"
"in.insulation_roof"
"in.insulation_slab"
"in.insulation_wall"
"in.interior_shading"
"in.iso_rto_region"
"in.lighting"
"in.lighting_interior_use"
"in.misc_extra_refrigerator"
"in.misc_freezer"
"in.misc_gas_fireplace"
"in.misc_gas_grill"
"in.misc_gas_lighting"
"in.misc_hot_tub_spa"
"in.misc_pool"
"in.misc_pool_heater"
"in.misc_pool_pump"
"in.misc_well_pump"
"in.natural_ventilation"
"in.neighbors"
"in.occupants"
"in.orientation"
"in.overhangs"
"in.plug_load_diversity"
"in.plug_loads"
"in.puma"
"in.puma_metro_status"
"in.pv_orientation"
"in.pv_system_size"
"in.radiant_barrier"
"in.range_spot_vent_hour"
"in.reeds_balancing_area"
"in.refrigerator"
"in.roof_material"
"in.schedules"
"in.solar_hot_water"
"in.state"
"in.tenure"
"in.units_represented"
"in.usage_level"
"in.vacancy_status"
"in.vintage"
"in.vintage_acs"
"in.water_heater_efficiency"
"in.water_heater_fuel"
"in.water_heater_in_unit"
"in.weather_file_city"
"in.weather_file_latitude"
"in.weather_file_longitude"
"in.window_areas"
"in.windows"
"upgrade.insulation_roof"
"upgrade.water_heater_efficiency"
"upgrade.hvac_cooling_efficiency"
"upgrade.infiltration_reduction"
"upgrade.geometry_foundation_type"
"upgrade.clothes_dryer"
"upgrade.insulation_ceiling"
“upgrade.ducts"
"upgrade.hvac_heating_type"
"upgrade.insulation_wall"
"upgrade.insulation_foundation_wall"
"upgrade.hvac_heating_efficiency"
"upgrade.cooking_range"
"Dry.Bulb.Temperature...C."
"Relative.Humidity...."

"Wind.Speed..m.s."
"Wind.Direction..Deg."
"Global.Horizontal.Radiation..W.m2."
"Direct.Normal.Radiation..W.m2."
"Diffuse.Horizontal.Radiation..W.m2."
"out.electricity.ceiling_fan.energy_consumption_total"
"out.electricity.clothes_dryer.energy_consumption_total"
"out.electricity.clothes_washer.energy_consumption_total"
"out.electricity.cooling_fans_pumps.energy_consumption_total"
"out.electricity.cooling.energy_consumption_total"
"out.electricity.dishwasher.energy_consumption_total"
"out.electricity.freezer.energy_consumption_total"
“out.electricity.heating_fans_pumps.energy_consumption_total"
"out.electricity.heating_hp_bkup.energy_consumption_total"
"out.electricity.heating.energy_consumption_total"
"out.electricity.hot_tub_heater.energy_consumption_total"
"out.electricity.hot_tub_pump.energy_consumption_total"
"out.electricity.hot_water.energy_consumption_total"
"out.electricity.lighting_exterior.energy_consumption_total"
"out.electricity.lighting_garage.energy_consumption_total"
"out.electricity.lighting_interior.energy_consumption_total"
"out.electricity.mech_vent.energy_consumption_total"
"out.electricity.plug_loads.energy_consumption_total"
"out.electricity.pool_heater.energy_consumption_total"
"out.electricity.pool_pump.energy_consumption_total"
"out.electricity.pv.energy_consumption_total"
"out.electricity.range_oven.energy_consumption_total"
"out.electricity.refrigerator.energy_consumption_total"
"out.electricity.well_pump.energy_consumption_total"
"hour_total_electricity_usage"
"month_total_electricity_usage"

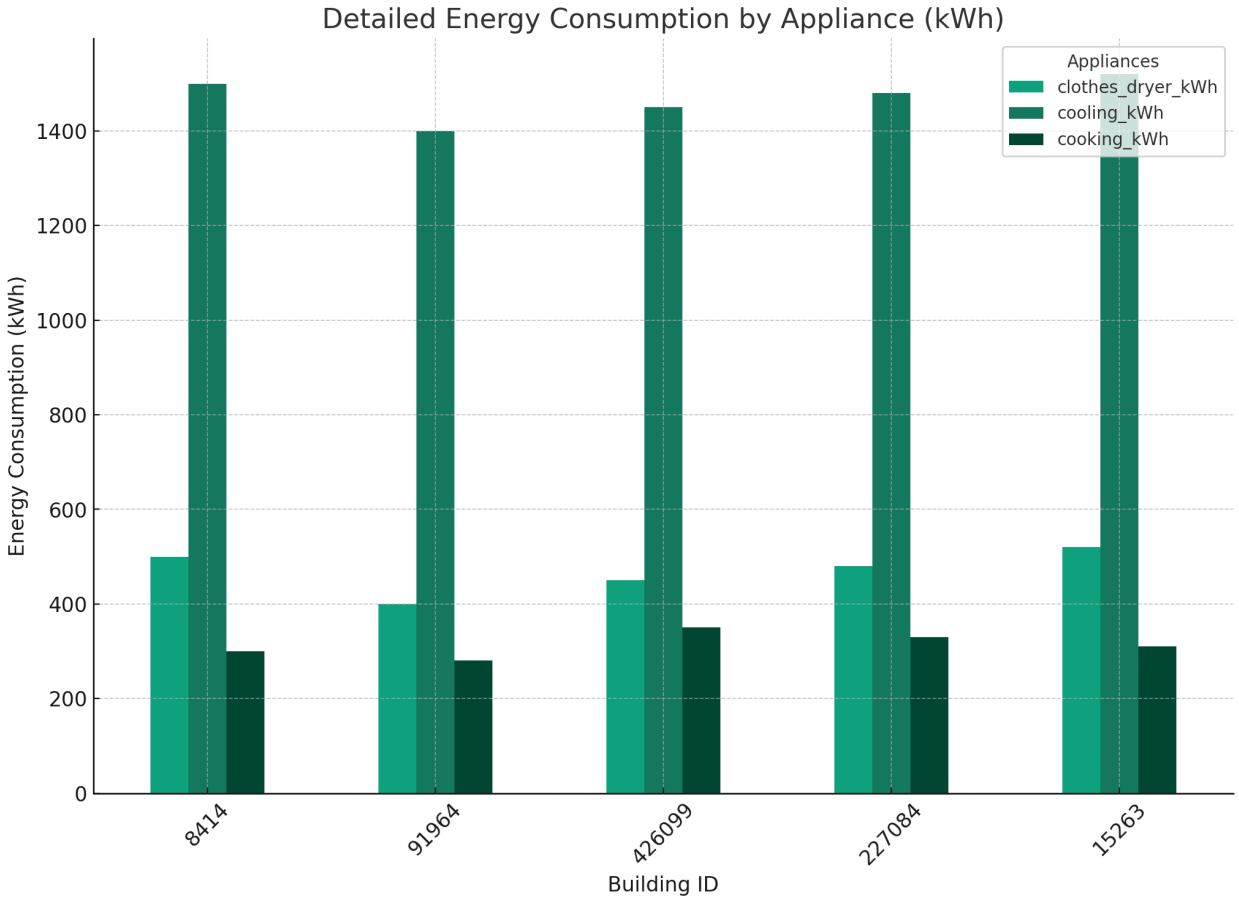
Prediction Models

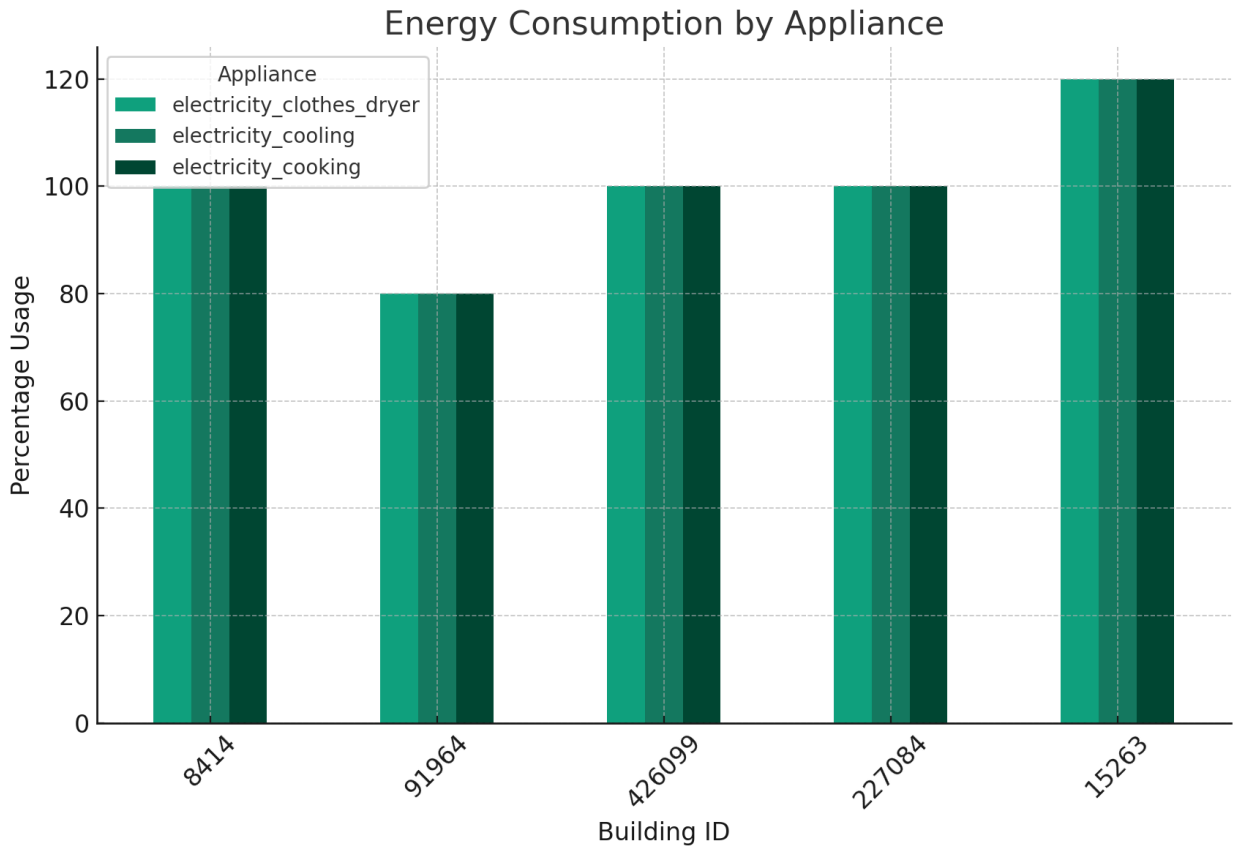
Model 1: Basic Model

- Variables:
  - "time"
  - "bldg_id"
  - "In.sqft"
  - "In.bedrooms"
  - "In.bathroom_spot_vent_hour"
  - "In.building_america_climate_zone"
  - "In.geometry_floor_area"
  - In.geometry_stories"
  - "In.geometry_wall_type"
  - "In.heating_setpoint"
  - "In.cooling_setpoint"
  - "In.hvac_heating_type"
  - "in.hvac_cooling_type"
- Rationale:
  - Start with essential building information and time-related variables.

Model 2: Appliance Usage Model

- Variables:
  - Appliance energy consumption variables (e.g., "out.electricity.clothes_dryer.energy_consumption", "out.electricity.cooling.energy_consumption", etc.)
  - "time"
  - "bldg_id"
- Rationale:
  - Focus on the energy consumption of different appliances and their impact on total energy usage.


Detailed Energy Consumption by Appliance (kWh)
Appliances
clothes_dryer_kWh
cooling_kWh
cooking_kWh
Energy Consumption (kWh)
0
200
400
600
800
1000
1200
1400
8414
91964
426099
227084
15263
Building ID



Model 3: Weather-Related Model

- Variables:
  - Weather variables (e.g., "Dry.Bulb.Temperature...C.", "Relative.Humidity....", "Wind.Speed..m.s.", etc.)
  - "time"
  - "bldg_id"
- Rationale:
  - Explore the influence of weather conditions on energy usage.

Model 5: Comprehensive Model

- Variables:
  - A combination of variables from the previous models, including building characteristics, appliance usage, weather conditions, etc.
- Rationale:
  - Build a comprehensive model that considers multiple factors influencing energy usage.